



NATIONAL 5 ENGINEERING SCIENCE · ENGINEERING CONTEXTS

Engineering Roles and Disciplines

Course notes

Version: 26 September 2026

Name _____

Class _____

Learning outcomes

Tick each box when you feel confident you can do it. Come back to this page before your unit test.

BY THE END OF THIS TOPIC...

✓

- | | | |
|---|---|--------------------------|
| 1 | I can describe what engineers do and why engineering matters to society. | ☐ |
| 2 | I can describe the work of civil, structural, mechanical, electrical, electronic, chemical and environmental engineers. | ☐ |
| 3 | I can give examples of projects that each branch of engineering works on. | ☐ |
| 4 | I can identify the branch of engineering responsible for a given task in a project. | ☐ |
| 5 | I can explain how engineers from different disciplines work together on a large project. | ☐ |
| 6 | I can describe the roles of engineers at each stage of a project: designing, implementing, testing and operating. | ☐ |
| 7 | I can explain why engineers follow a code of conduct, and name a professional engineering body. | ☐ |
| 8 | I can describe the qualifications and career paths that lead to a job in engineering. <i>(not examined)</i> | ☐ |
| 9 | I can describe specific roles that engineers would carry out in a given context. | ☐ |

Where to write your answers: if you have access to a device, use the online course notes — you can type your answers into the answer panels and they are saved as you go. Otherwise, write your answers in your jotter, with the task and question number beside each answer.



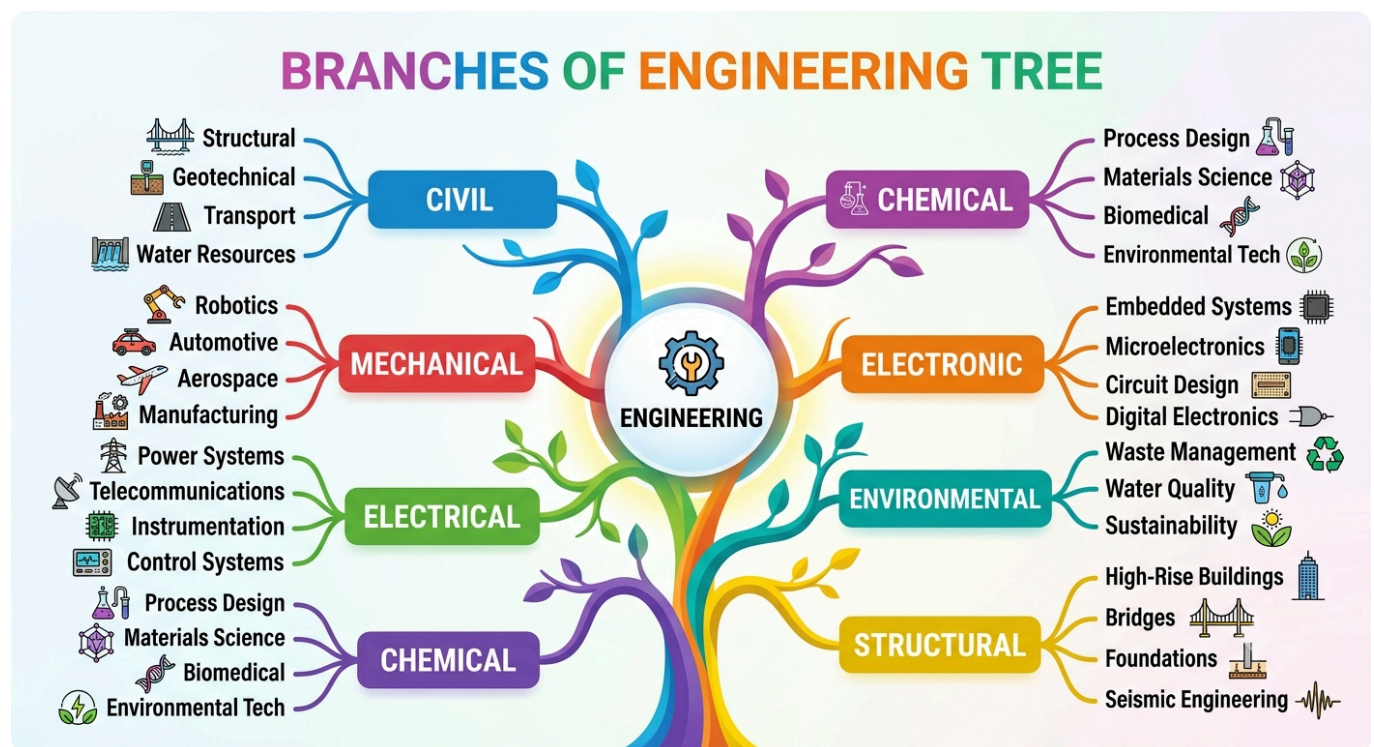
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Section 1: What is Engineering?

Engineering is the application of **science, mathematics and technology** to design and build solutions to real-world problems. Engineers improve the world around us — from the bridges we cross, to the phones in our pockets, to the medicines that keep us healthy.

The work of engineers is everywhere. Without engineering there would be no clean water supply, no electricity grid, no roads, no internet and no modern medicine. Engineering shapes civilisation.

Engineers design, build, test and maintain systems and structures that improve people's lives.



Section 2: The Seven Main Branches of Engineering

The QS specification requires you to know seven main branches. Each branch has a different focus, but they frequently **work together** on the same project.

Civil Engineering

Definition: Planning, designing, building and maintaining the infrastructure of society.

Examples of work: Roads, bridges, tunnels, dams, airports, water treatment plants, sewage systems, railway lines.

Real project example: Designing and implementing the deep underground tunnels and water routing for the Shieldhall Tunnel in Glasgow, ensuring they integrate safely into the city's complex drainage and flood management system.

Structural Engineering

Definition: Designing structures so they can safely support loads and resist forces without failing.

Examples of work: Load-bearing frames of buildings, bridges, towers, offshore platforms.

Real project example: Calculating wind forces and testing the load-bearing strength of the internal steel framework of The Kelpies, ensuring safe implementation so the 30-metre sculptures withstand severe storms without failing.

Mechanical Engineering

Definition: Designing systems involving motion, force and energy — essentially anything that moves or has moving parts.

Examples of work: Engines, turbines, manufacturing machinery, robots, HVAC systems, prosthetic limbs.

Real project example: Designing, testing, and controlling the complex mechanical drive gears and electric motors that rotate The Falkirk Wheel to lift and transfer canal boats between two waterways.

Electrical Engineering

Definition: Generating, distributing and using electrical power; designing electrical systems and devices.

Examples of work: Power stations, national grid, electric motors, transformers, lighting systems, wind turbines.

Real project example: Designing and implementing the high-voltage electrical cabling and substations for Whitelee Windfarm, solving the challenge of transmitting massive amounts of renewable power into the national grid.

Electronic Engineering

Definition: Designing circuits, devices and systems that process electrical signals for control and communication.

Examples of work: Smartphones, computers, control systems, telecommunications, medical imaging equipment, sensors.

Real project example: Designing and testing the sensitive electronic circuits and sensor systems for hospital MRI

Chemical Engineering

Definition: Designing industrial processes that use chemical, biological or physical transformations of raw materials.

Examples of work: Oil refining, pharmaceuticals, food processing, plastics, fertilisers, water treatment chemicals.

Real project example: Designing and testing chemical

scanners — technology originally pioneered in Scotland
— used to control complex magnetic imaging systems.

conversion and refining processes to turn industrial and
whisky distillery by-products into clean biofuels,
implementing greener alternatives to fossil fuels.

Environmental Engineering

Definition: Protecting the environment from the negative effects of human activity, and designing systems to restore and improve it.

Examples of work: Pollution control, waste management, air quality systems, ecological impact assessments, renewable energy planning.

Real project example: Testing underwater sensor data and implementing sonar tracking systems for the MeyGen tidal power project in the Pentland Firth, controlling the environmental impact to ensure the massive underwater turbines do not harm local marine life like seals and dolphins.

Task 1 — Branches of engineering

Complete the table below by writing a brief definition and two real-world examples of work for each branch of engineering.

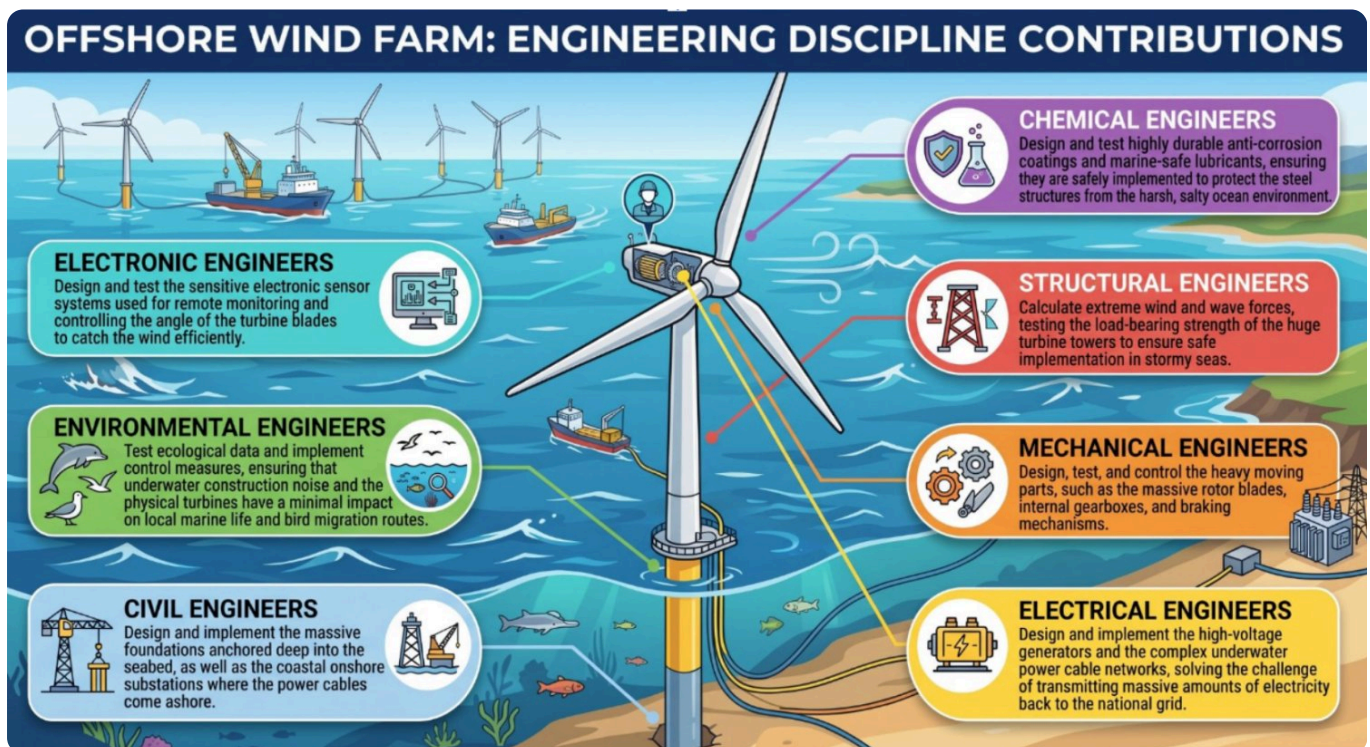
Branch	Definition (in your own words)	Two examples of projects or work
Civil		
Structural		
Mechanical		
Electrical		
Electronic		
Chemical		
Environmental		

Section 3: How Engineering Disciplines Work Together

In the real world, major engineering projects are never the work of just one discipline. They require **multiple branches of engineering working together** at every stage. This is called **interdisciplinary engineering**.

Consider building a new offshore wind farm. It needs:

- **Civil engineers** to design and implement the massive foundations anchored deep into the seabed, as well as the coastal onshore substations where the power cables come ashore
- **Structural engineers** to calculate extreme wind and wave forces, testing the load-bearing strength of the huge turbine towers to ensure safe implementation in stormy seas
- **Mechanical engineers** to design, test, and control the heavy moving parts, such as the massive rotor blades, internal gearboxes, and braking mechanisms
- **Electrical engineers** to design and implement the high-voltage generators and the complex underwater power cable networks, solving the challenge of transmitting massive amounts of electricity back to the national grid
- **Electronic engineers** to design and test the sensitive electronic sensor systems used for remote monitoring and controlling the angle of the turbine blades to catch the wind efficiently
- **Chemical engineers** to design and test highly durable anti-corrosion coatings and marine-safe lubricants, ensuring they are safely implemented to protect the steel structures from the harsh, salty ocean environment
- **Environmental engineers** to test ecological data and implement control measures, ensuring that underwater construction noise and the physical turbines have a minimal impact on local marine life and bird migration routes, and to monitor the long-term impact of the project on the marine environment



Task 2 — Integrated engineering project

Choose an engineering project (e.g. a new bridge, sports stadium, hospital, electric vehicle, or a project of your choice). For each branch of engineering, describe how it contributed to the project.

My project is:

Complete the table to show how each engineering branch contributed to your chosen project.

Branch	Contribution to the project
Civil	
Structural	
Mechanical	
Electrical	
Electronic	
Chemical	
Environmental	

Write a short paragraph explaining how the different branches of engineering worked **together** to ensure the success of your project.

Section 4: Roles of Engineers

Within any engineering project, engineers take on different **roles** at different stages. The QS specification requires you to know four key stages and what engineers do at each.

Stage	What engineers do	Examples
 Designing	Calculate, simulate, select materials, research, survey, produce drawings and specifications	Calculating the load on a bridge beam; simulating airflow over a wind turbine blade; selecting the correct steel grade for a structure
 Implementing	Organise teams, monitor progress, troubleshoot problems, liaise with other engineers and contractors	Managing construction workers; solving unexpected ground conditions; coordinating with electrical and civil teams on a wind farm
 Testing	Monitor sensors, analyse test results, adapt original designs based on findings	Load testing a bridge; checking electrical systems under full power; stress testing a new aircraft component
 Controlling	Monitor ongoing performance, maintain systems, respond to faults, ensure safe operation	Monitoring the National Grid frequency; controlling traffic signals; managing the automated monitoring systems on the Queensferry Crossing

ENGINEERING DESIGN PROCESS — CIRCULAR FLOW



Professional bodies and codes of conduct: Most engineers belong to a professional body for their discipline, such as the **ICE** (Institution of Civil Engineers), **IStructE** (Institution of Structural Engineers), **IMechE** (Institution of Mechanical Engineers), **IET** (Institution of Engineering and Technology) or **IChemE** (Institution of Chemical Engineers). Every member must follow that body's **code of conduct**: a set of rules for behaving professionally and responsibly. Typical rules include:

- put the **health and safety of the public** first
- only take on work you are **competent** to do
- be **honest** and accurate — never hide faults, risks or poor test results
- protect the **environment** and act in the public interest
- keep your knowledge and skills **up to date** throughout your career

Codes of conduct matter because people's lives depend on engineers' work. If a bridge, lift or power system is designed carelessly, or a known problem is covered up, people can be seriously hurt. An engineer who breaks the code can be disciplined or lose their professional status.

Task 3 — Roles of engineers

Part A: Identifying Key Roles — think about your chosen project from Task 2 (or a major structural project like the retractable roof on Centre Court at Wimbledon). A project of this scale requires teamwork, but not every type of engineer is needed for every project. Identify two engineering disciplines that were absolutely essential to your chosen project's success.

Discipline 1 (e.g. Structural Engineer):

Discipline 2 (e.g. Mechanical Engineer):

Part B: The Project Lifecycle — for your two chosen disciplines, briefly describe their specific responsibilities during the three main phases of the project.

	Discipline 1	Discipline 2
1. Designing What did they calculate, model, draw or plan before construction began?		
2. Implementing What was their role while the project was actively being built or manufactured?		
3. Testing & Controlling How did they ensure safety, functionality and quality once the physical work was completed or underway?		

Explain why engineers must follow a professional code of conduct. Name one professional engineering body and one rule from its code of conduct.

Section 5: Career Paths in Engineering

This section is not directly assessed as part of the QS specification, but it's well worth exploring — it connects your course work to real career and Developing the Young Workforce (DYW) opportunities.

Engineering offers a wide range of career options at different qualification levels:

- **Technician** — practical hands-on work, often HNC/HND qualified
- **Incorporated Engineer (IEng)** — degree-qualified, applies established techniques
- **Chartered Engineer (CEng)** — highest professional status, leads innovation and complex projects

The pathway from school to Chartered Engineer typically involves: National 5 → Higher → Advanced Higher → Degree (BEng or MEng) → Graduate experience → Chartership via professional body (e.g. IET for electrical/electronic, IMechE for mechanical, ICE for civil).

Engineering is one of the best-paid and most in-demand careers in Scotland and the UK. Scotland has a particularly strong demand for engineers in renewable energy, civil infrastructure, oil and gas, and advanced manufacturing.

Task 4 — Research task

Research a branch of engineering that interests you. Find out: what engineers in this field do, a specific real-world project they have worked on, the qualifications needed, and one engineering challenge currently facing this discipline.

Section 6: Engineering Challenges

Tasks 1–4 checked that you know the seven branches of engineering and the four roles engineers carry out. The three challenges below work differently. Each one gives you a real-world context and a written **specification**, the same way an assignment brief does. Nobody tells you which engineers are needed: you have to work that out from the specification and justify your choices.

How to approach these: Read every specification point and ask “*what kind of problem is this?*” Loads and forces point to structural engineers; moving parts to mechanical; generating or transmitting power to electrical; sensors, monitoring and control to electronic; materials, coatings and processes to chemical; wildlife, pollution and waste to environmental; ground works, water and transport routes to civil. Most points need more than one discipline, so look for where they overlap.

Challenge 1 — Community hydro scheme

A community in a Highland glen wants to build a small hydro-electric scheme. Water will be taken from a burn high on the hillside, carried downhill in a buried pipe, and used to spin a turbine in a small powerhouse beside the glen road. The river below the scheme is an important spawning ground for salmon.

-  The scheme must generate enough electricity to supply around 60 homes and feed it into the local grid.
-  The intake on the burn must always leave enough water flowing to protect fish and other wildlife.
-  The pipe must be buried along its full length so that it can't be seen from the glen road.
-  The scheme must be monitored and controlled remotely, as nobody will be on site from day to day.

Challenge 1 — Community hydro scheme

C1a. For each point of the specification, identify the branch of engineering that would take the lead and explain why.

C1b. Choose two of the disciplines you named in (a). For each one, describe one thing it would do during the **designing** stage and one thing it would do during the **implementing** stage.

C1c. Specification points (i) and (iv) can't be met by one discipline working alone. Choose one of these points and explain how two disciplines would need to work together to meet it.

C1d. Once the scheme is running, describe how engineers would **test** and **control** it to make sure it keeps meeting specification point (ii).

Challenge 2 — Step-free station footbridge

A town's railway station has an old stepped footbridge that people using wheelchairs, pushchairs or heavy luggage can't use. It is to be replaced with a new step-free footbridge with a lift tower on each platform. The railway underneath is electrified, with overhead power lines, and trains run every few minutes during the day.

-  The bridge must safely carry large crowds of passengers, including at busy times after football matches and concerts.
-  Each platform must have a lift big enough to carry a wheelchair user and a companion.
-  Construction must be planned so that trains are disrupted for as little time as possible.
-  The lifts must report faults automatically so that they can be repaired quickly.

Challenge 2 — Step-free station footbridge

C2a. Identify four branches of engineering needed on this project. For each one, state which specification point it is mainly responsible for and describe its contribution.

C2b. During construction, the team discovers that the ground under one of the lift towers is much softer than the original survey showed. Identify which discipline(s) would deal with this problem and describe what they would need to do.

C2c. Describe two tests that engineers should carry out on the lifts before the bridge opens to the public. For each, explain what the test is checking.

C2d. Suggest one improvement that would make the new footbridge better than the specification requires. Identify which discipline would be responsible for it.

Challenge 3 — Electric island ferry

A council plans to replace the diesel ferry on a short crossing between the mainland and a west-coast island with a battery-electric ferry. The ferry will be charged from the grid at the mainland pier during the few minutes it spends there loading and unloading vehicles.

- i The ferry's batteries must be recharged during its turnaround time at the mainland pier.
- ii The pier must be strengthened to hold the charging equipment and to withstand winter storms.
- iii The ferry's hull and the pier equipment must resist corrosion from seawater.
- iv The charge level and temperature of the batteries must be monitored continuously on every crossing.
- v Construction work at the pier must not pollute the harbour or harm marine wildlife.

Challenge 3 — Electric island ferry

C3a. Match each point of the specification to the branch of engineering most responsible for it. Then state whether any of the seven branches from Section 2 is **not** needed on this project, and justify your answer.

C3b. For specification point (iv), describe what engineers would do at each of the four stages.

Stage	What engineers would do for point (iv)
 Designing	
 Implementing	
 Testing	
 Controlling	

C3c. Explain one way that decisions made by the electrical engineers could affect the work of the civil or structural engineers on this project.

C3d. Describe one benefit and one drawback of replacing the diesel ferry with an electric one. (You'll look at impacts like these in much more detail in the next topic, Impacts of Engineering.)